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1.

```
#include<stdio.h>
int main(){
    printf("Enter The number of elements you want\n");
    int len = 0;
    scanf("%d", &len);
    int a[len];
    int out[len][2];
    int temp;
    int alraedy_present = 0;
    int ele = 0;

    printf("Enter Values: \n");
    for(int i = 0; i < len; i++){
        scanf("%d", &a[i]);
    }
    for(int i = 0; i < len; i++){
        temp = a[i];
        for(int j= 0; j < len;j++){
            if(temp == out[j][0]){
                alraedy_present = 1;
                out[j][1] += 1;
                break;
            }
        }
        if(!alraedy_present){
            out[ele][0] = temp;
            out[ele][1] = 1;
            ele++;
        }
        alraedy_present = 0;
    }
    for(int i = 0; i < ele; i++){
        printf("%d Occours: %d times\n", out[i][0], out[i][1]);
    }
    return 0;
}
```

```
neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./1.out
Enter The number of elements you want
8
Enter Values:
1
1
23
34
34
34
5
6
1 Occours: 2 times
23 Occours: 1 times
34 Occours: 3 times
5 Occours: 1 times
6 Occours: 1 times
```

2.

```
#include<stdio.h>
int main(){
    printf("Enter The number of elements you want\n");
    int len;
    scanf("%d", &len);
    int a[len];
    int out[len];
    int temp;
    int alraedy_present = 0;
    int ele = 0;

    printf("Enter Values: \n");
    for(int i = 0; i < len; i++){
```

```

        scanf("%d", &a[i]);
    }
    for(int i = 0; i < len; i++){
        temp = a[i];
        for(int j= 0; j < len;j++){
            if(temp == out[j]){
                alraedy_present = 1;
                break;
            }
        }
        if(!alraedy_present){
            out[ele] = temp;
            ele++;
        }
        alraedy_present = 0;
    }
    printf("The array after removing duplicates is: \n");
    for(int i = 0; i < ele; i++){

        printf("%d\n", out[i]);

    }
    return 0;
}

```

```

neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./2.out
Enter The number of elements you want
8
Enter Values:
1
1
2
3
3
3
4
5
The array after removing duplicates is:
1
2
3
4
5

```

3.

```
#include<stdio.h>
int main(){
    printf("Enter the number of rows and columns you require: \n");
    int r,c;
    r = 0;
    c = 0;
    scanf("%d %d", &r, &c);
    printf("Enter the values for Array A row wise: \n");
    int a[r][c], b[r][c], x[r][c];
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            scanf("%d", &a[i][j]);
        }
    }
    printf("Enter the values for Array B row wise: \n");
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            scanf("%d", &b[i][j]);
        }
    }
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            x[i][j] = a[i][j] + b[i][j];
        }
    }
    printf("The Matrix Sum is: \n");
    for(int i = 0; i < r; i++){
        for(int j = 0; j < c; j++){
            printf("%d ", x[i][j]);
        }
        printf("\n");
    }
}
```

```

neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./3.out
Enter the number of rows and columns you require:
2 2
Enter the values for Array A row wise:
1
2
3
4
Enter the values for Array B row wise:
6
7
8
9
The Matrix Sum is:
7 9
11 13

```

4.

```

#include<stdio.h>
int main(){
    printf("Enter the order of the matrix \n");
    int r;
    scanf("%d", &r);
    printf("Enter the values for Array A row wise: \n");
    int a[r][r];
    int p_dig =0, s_dig = 0;

    int x = 0, y = r-1;
    for(int i = 0; i < r; i++){
        for(int j = 0; j < r; j++){
            scanf("%d", &a[i][j]);
        }
    }
    for(int i = 0; i < r; i++){
        p_dig += a[i][i];
    }

    for(int i = 0; i < r; i++){
        s_dig += a[x][y];
        x++;
        y--;
    }

    printf("The Primary diagonal Sum is: %d\n",p_dig);
    printf("The Secondary diagonal Sum is: %d\n",s_dig);
}

```

```
neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./4.out
Enter the order of the matrix
2
Enter the values for Array A row wise:
2
1
3
5
The Primary diagonal Sum is: 7
The Secondary diagonal Sum is: 4
```